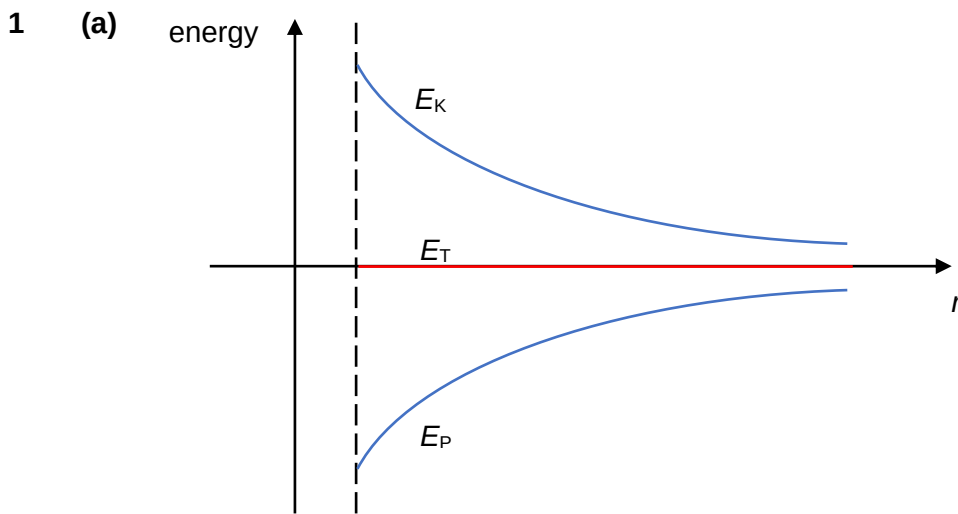




E_K positive and decreasing, not touching x-axis

B1

E_P (roughly, but not too far off) symmetrical to E_K

B1

E_T clearly shown (visibly or written in text) on the x-axis

B1

- (b) (i) Kinetic energy lost equals the gain in gravitational potential energy when object of mass m moves from the surface of Earth to infinity
 $\frac{1}{2} mv^2 - 0 = 0 - (-GMm / R)$

B1

$$v^2 = 2GM / R$$

- (ii) gravitational field strength at the surface of planet $g = GM / R^2$ equals the acceleration of free fall at the surface of planet

B1

from (i), $v^2 = 2 (GM / R^2) \times R$

M1

$$v^2 = 2gR$$

A0

- (c) (i) $\frac{3}{2} kT = \frac{1}{2} m v^2 = mgR$

C1

$$T = 2mgR / 3k = (2 \times 6.6 \times 10^{-27} \times 9.81 \times 6.4 \times 10^6) / (3 \times 1.38 \times 10^{23})$$

C1

$$T = 2.0 \times 10^4 \text{ K}$$

A1

- (ii) temperature where all substances have minimum internal energy

B1

2

2 (a) sum of random distribution of kinetic and potential energies of the molecules of the gas R1